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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Clinical data were obtained from institutional medical records and pathology databases of participating centers. Preoperative contrast-enhanced CT images were retrieved from institutional picture archiving and communication systems. Pathology H&E images were obtained from institutional pathology slide-scanning systems and the TCGA-STAD collection in The Cancer Imaging Archive. RNA-sequencing libraries were prepared by Genedenovo Biotechnology Co., Ltd. and sequenced on the Illumina NovaSeq X Plus platform.
Data analysis	The custom code used for computational analysis in this study has been deposited in a public GitHub repository at https://github.com/hebeidpa/GC-MLM-RCSA. All experiments and analyses were conducted using Python v3.8.10, R v4.3.3, and IBM SPSS Statistics v28.0. The computational workflow was implemented using commonly used open-source libraries for medical image analysis, pathology image analysis and machine learning, including PyTorch v2.1.2, torchvision v0.16.2, PyTorch Lightning v2.2.1, timm v0.9.8, torchmetrics v0.7.3, NumPy v1.26.4, SimpleITK v2.2.1 or later, scikit-image v0.19.3 or later, acvl-utils v0.2.3 or later and <v0.3, dynamic-network-architectures v0.3.1 or later and <v0.4, batchgenerators v0.25.1 or later, batchgeneratorsv2 v0.2 or later, open_clip_torch v0.2.1 or later, webdataset v0.2.31 or later, blosc2 v3.0.0b1 or later, pandas, SciPy, scikit-learn, scikit-survival, matplotlib, seaborn, OpenCV-Python, nibabel, h5py, pyarrow, ruamel.yaml, safetensors, transformers, sentencepiece, tqdm, Pillow, einops, accelerate and fairscale. R-based statistical and bioinformatic analyses used glmnet, rms, pROC, survival, survminer, edgeR, limma, GSVA, ESTIMATE, MCPcounter and EcoTyper v1.0. RNA-sequencing data were processed using fastp, HISAT2, StringTie and RSEM. CIBERSORT was performed with 1,000 permutations.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The pathology H&E images and CT images from the TCGA-STAD collection used in this study are available from The Cancer Imaging Archive (https://www.cancerimagingarchive.net/collection/tcga-stad/). The raw RNA-sequencing data generated in this study have been deposited in the Genome Sequence Archive for Human (GSA-Human) at the National Genomics Data Center (NGDC) under accession code HRA013404 (https://ngdc.cncb.ac.cn/gsa-human/browse/HRA013404). Access to these data is controlled due to patient privacy considerations and regulatory requirements. Qualified researchers may request access through the GSA-Human Data Access Committee, and will be required to submit a brief research proposal and complete a standard application form describing the project, the data requested, and the intended use, in accordance with institutional ethical guidelines and applicable regulations. Participant-level clinical, CT and pathology image data are available under restricted access due to patient privacy laws and institutional review board requirements; access can be obtained by submitting a formal research proposal to Qun Zhao (zhaoqun@hebmu.edu.cn). Requests will be evaluated by the institutional data management committee based on ethical compliance and scientific merit within 14 days, and once granted, access will be available for the duration of the approved research project. The processed sequencing data are available at GSA-Human. Source data are provided with this paper.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Sex was recorded as part of routine clinical data and was self-reported by participants. Analyses were stratified by sex where relevant. No gender-specific analyses were performed beyond biological sex differentiation.
Reporting on race, ethnicity, or other socially relevant groupings	Race or ethnicity data were not collected or analyzed in this study.
Population characteristics	The study population comprised patients with locally advanced gastric cancer who had available preoperative CT images, haematoxylin and eosin-stained pathological slides, clinicopathological variables and follow-up data. Covariate-relevant characteristics included age, sex, ECOG performance status, tumour location, histological subtype, Lauren classification, pathological tumour–node–metastasis stage, treatment information, molecular markers including HER2 and PD-L1 where available, and recurrence or survival outcomes. Detailed baseline characteristics are provided in the main and supplementary tables.
Recruitment	We retrospectively analyzed data from patients with LAGC treated between January 2012 and December 2019 at four medical centers in Hebei Province: the Fourth Hospital of Hebei Medical University (FHHMU), Shijiazhuang People's Hospital (SJZPH), Baoding Central Hospital (BDCH), and Hengshui People's Hospital (HSPH). Data from two additional centers in southern China, Renmin Hospital of Wuhan University (WHRH) and Nanjing Jinling Hospital (NJJLH), were also included for external validation.
Ethics oversight	The study was approved by the Ethics Committee of the Fourth Hospital of Hebei Medical University (approval number 2025KT150), which acted as the central ethics committee; all participating centres accepted this approval. Owing to the retrospective design and use of de-identified data, informed consent was waived. The waiver of written informed consent was granted by the ethics committee on the basis that the study met all of the following criteria: (1) the research involved no more than minimal risk to participants, as it used only pre-existing, routinely collected clinical, imaging, and pathology data; (2) the waiver would not adversely affect the rights and welfare of the participants; (3) the research could not practicably be carried out without the waiver, given the retrospective, multicentre design, the large number of eligible patients accrued over more than a decade, and the fact that a substantial proportion of patients had died or become untraceable at the time of analysis; and (4) all records were fully de-identified before analysis, and no attempt was made to re-identify individual participants.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical method was used to predetermine sample size. For the retrospective cohorts, all consecutive eligible patients who met the predefined inclusion and exclusion criteria and had complete clinical, CT, pathological and follow-up data during the study period were included. Although no a priori calculation was performed, a post-hoc check using the criteria of Riley et al. confirmed the adequacy of the sample size: for the final RCSA model (the radiomic and pathomic signatures entered as single composite predictors plus three independent clinical predictors), an outcome proportion of 0.166 (128 events among 770 patients) and a conservative Cox-Snell R^2 of 0.15 yielded a minimum required size of 384 (18.3 events per predictor parameter), which the development cohort exceeded. Independent internal, external, prospective and public-data cohorts were used to evaluate model robustness and generalizability. For RNA-sequencing and immune-microenvironment analyses, 82 tumour specimens with available high-quality RNA-sequencing data were included; this sample size was determined by the availability of prospectively collected specimens that met sequencing quality-control requirements and was considered sufficient for exploratory transcriptomic and immune-profiling analyses.
Data exclusions	(1) Receipt of neoadjuvant chemotherapy, targeted therapy, or immunotherapy before surgery; (2) surgery-related death (within 90 days postoperatively); (3) Loss to follow-up; (4) CT image artifacts or missing CT images; (5) Absence of pathological hematoxylin and eosin (H&E) images; (6) Missing clinical information
Replication	Each patient contributed one independent clinical, imaging, pathological and, where applicable, tumour RNA-sequencing sample. Patient-level observations were treated as independent biological replicates, and no technical replicate was treated as an independent observation. Model performance was assessed using tenfold cross-validation in the training cohort and was further evaluated in independent internal validation, external validation, prospective validation and public-data cohorts. The RNA-sequencing samples were processed and sequenced in a single batch to reduce technical batch effects.
Randomization	As this was a retrospective observational study, no random allocation was performed.
Blinding	Given the retrospective and observational nature of the study, formal blinding was not applicable to clinical data extraction. Image segmentation, feature extraction and model evaluation were conducted according to standardized protocols. Outcome labels were not used during radiomic and pathomic feature extraction.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☒	☐	Antibodies
☒	☐	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☒	☐	Animals and other organisms
☐	☒	Clinical data
☒	☐	Dual use research of concern
☒	☐	Plants

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	This study includes an exploratory post-hoc analysis of data from a registered prospective multicenter clinical trial: NCT02555358. The primary retrospective cohort was not registered as a clinical trial, as it is a retrospective observational analysis
Study protocol	No formal trial protocol was registered. The study followed an internal IRB-approved protocol, which is available from the corresponding author upon reasonable request.
Data collection	Data were obtained from institutional medical records and pathology databases in accordance with ethical approval. All data were de-identified prior to analysis
Outcomes	The primary endpoint of this study was the occurrence of postoperative MLM, and the secondary endpoint was OS.

Plants

Seed stocks	N/A
Novel plant genotypes	NA
Authentication	NA